3.

$e^{(t)} = x^{(t)}L$, $h^{(t)} = \text{Sigmoid}(h^{(t-1)}H + e^{(t)}I + b_1)$

$\hat{y}^{(t)} = \text{Softmax}(h^{(t)}U + b_2)$.

$L \in \mathbb{R}^{|V| \times d}$ $H \in \mathbb{R}^{D_h \times D_h}$, $I \in \mathbb{R}^{d \times D_h}$, $b_1 \in \mathbb{R}^{D_h}$, $U \in \mathbb{R}^{D_h \times |V|}$, $b_2 \in \mathbb{R}^{|V|}$.

(a).

$y_k^{(t)}$가 $y^{(t)}$의 유일한 nonzero element라고 하자. 그러면,

$CE(y^{(t)}, \hat{y}^{(t)}) = \cancel{-\sum y_j^{(t)} \hat{y}_j^{(t)} = \sum y_j^{(t)} \log \hat{y}_j^{(t)} = -\log y_k^{(t)} = \log \frac{1}{\hat{y}_k^{(t)}}}$

$= -\log \hat{y}_k^{(t)} = \log \frac{1}{\hat{y}_k^{(t)}}$ 이고,

$PP^{(t)}(y^{(t)}, \hat{y}^{(t)}) = \frac{1}{\hat{y}_k^{(t)}}$ 이므로, $CE = \log PP$ 가 성립해서

CE의 arithmetic mean을 최소화하는 것과 PP의 geometric mean을 최소화하는 것은 같은 의미가 된다.

i.e. $\frac{1}{N}\sum \log \frac{1}{y}$: (arithmetic) mean of CE.

$\cancel{\log(\prod(PP))^{\frac{1}{n}} = \frac{1}{n}\sum PP}$

$\prod \left(\frac{1}{y}\right)^{\frac{1}{n}}$: (geometric) mean of PP. ← 여기에 log를 취하면

$\log \prod \left(\frac{1}{y}\right)^{\frac{1}{n}} = \frac{1}{n}\sum \log \frac{1}{y}$ 이고, log는 증가함수이므로 같은 의미임.

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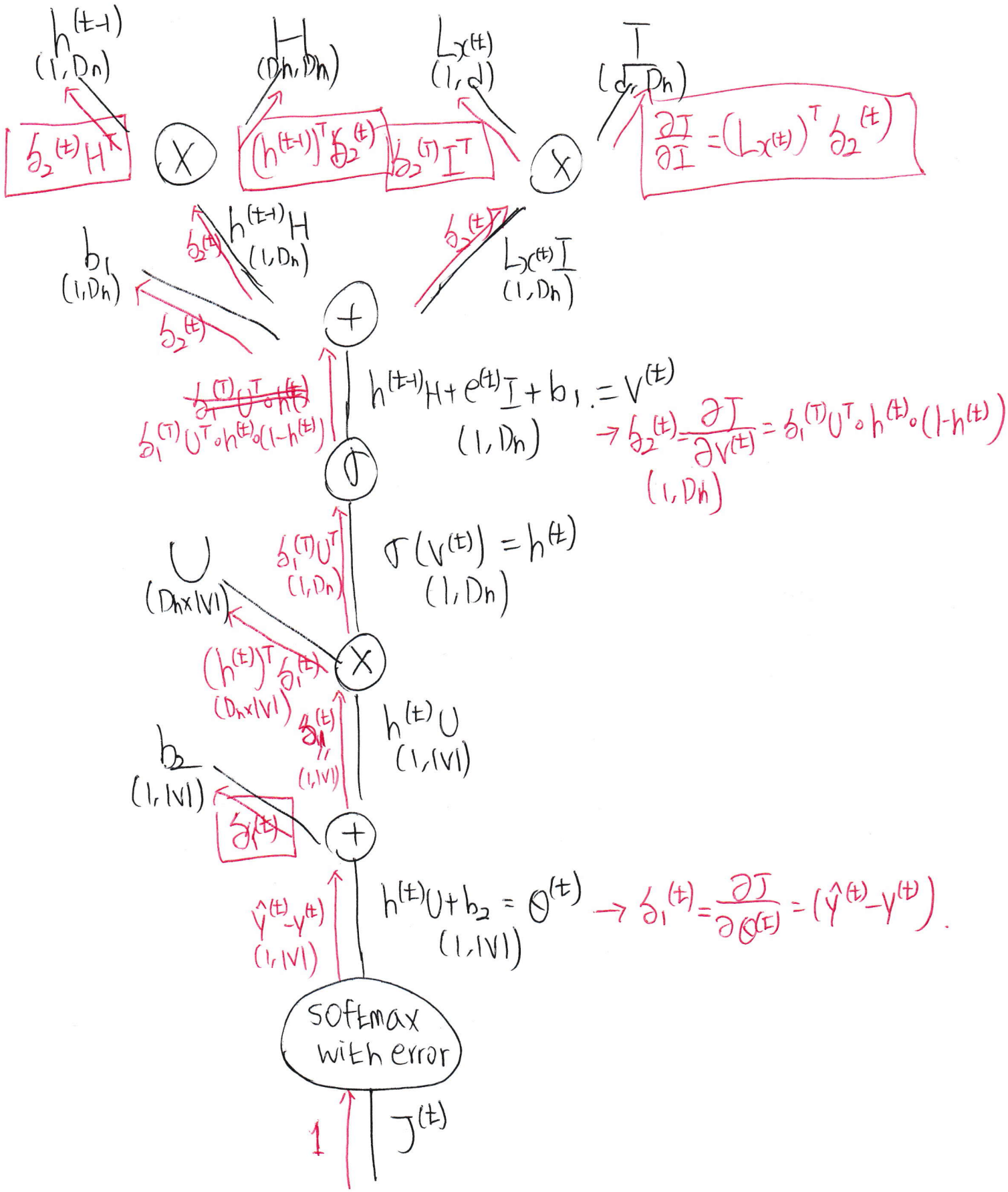
model prediction이 random하면 $\hat{y}^{(t)} = \frac{1}{|V|}$ 인 것과 같으므로

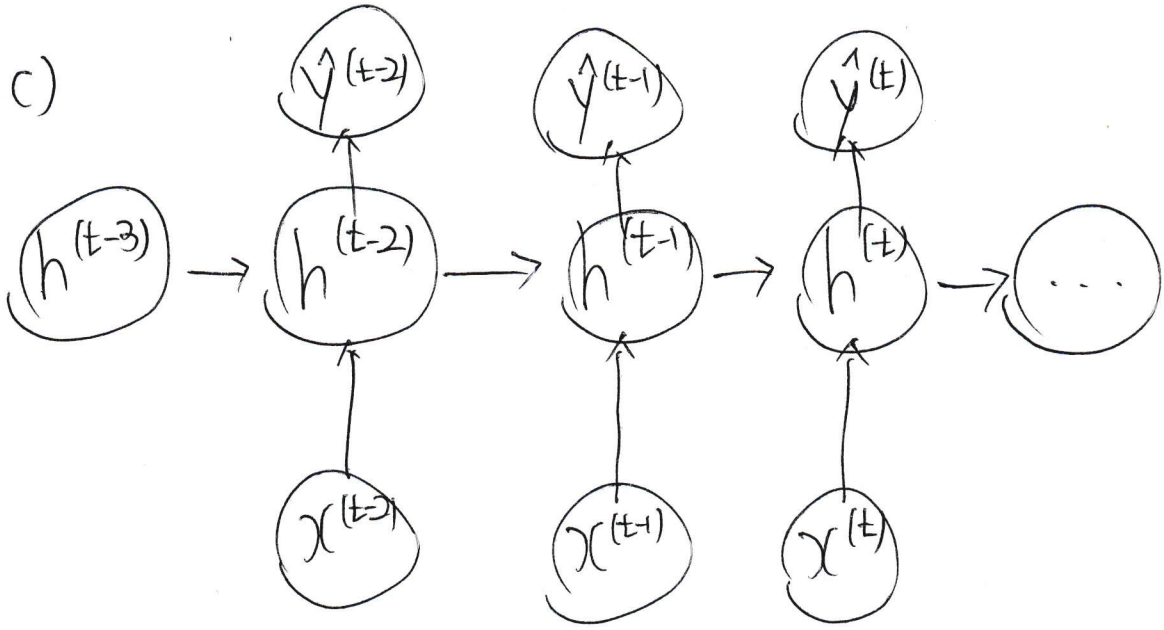
$CE = -\log \hat{y}^{(t)} = \log 10000$.

(b)

$x^{(t)}L$을 $L_{x^{(t)}}$로 쓰자. Computation graph를 그리면



(c)



① $$\frac{\partial J}{\partial L_{x^{(t-1)}}} = \frac{\partial J}{\partial h^{(t-1)}} \cdot \frac{\partial h^{(t-1)}}{\partial v^{(t-1)}} \cdot \frac{\partial v^{(t-1)}}{\partial L_{x^{(t-1)}}}$$

$$= \delta^{(t-1)} \cdot \sigma'(v^{(t-1)}) \cdot I^T.$$

여기서 $\sigma'(v^{(t-1)}) = \frac{\partial h^{(t-1)}}{\partial v^{(t-1)}} = \mathrm{diag}(h^{(t-1)} \circ (1-h^{(t-1)}))$인데, 그 이유는

$h^{(t-1)} = \sigma(v^{(t-1)}) = (\sigma(v_1^{(t-1)}), \sigma(v_2^{(t-1)}), \cdots, \sigma(v_{D_h}^{(t-1)}))$이고,

$$\frac{\partial h^{(t-1)}}{\partial v^{(t-1)}} = \begin{pmatrix} \frac{\partial h_1^{(t-1)}}{\partial v_1^{(t-1)}} & \cdots & \frac{\partial h_1^{(t-1)}}{\partial v_{D_h}^{(t-1)}} \\ \vdots & & \\ \frac{\partial h_{D_h}^{(t-1)}}{\partial v_1^{(t-1)}} & \cdots & \frac{\partial h_{D_h}^{(t-1)}}{\partial v_{D_h}^{(t-1)}} \end{pmatrix} = \begin{pmatrix} h_1^{(t-1)}(1-h_1^{(t-1)}) & & 0 \\ & \ddots & \\ 0 & & h_{D_h}^{(t-1)}(1-h_{D_h}^{(t-1)}) \end{pmatrix}$$

이기 때문임.

② $$\left.\frac{\partial J}{\partial I}\right|_{(t-1)} = \frac{\partial J}{\partial h^{(t-1)}} \cdot \frac{\partial h^{(t-1)}}{\partial v^{(t-1)}} \cdot \frac{\partial v^{(t-1)}}{\partial I} = (L_{x^{(t-1)}})^T \delta^{(t-1)} \sigma'(v^{(t-1)})$$

③ $$\left.\frac{\partial J}{\partial H}\right|_{(t-1)} = \frac{\partial J}{\partial h^{(t-1)}} \cdot \frac{\partial h^{(t-1)}}{\partial v^{(t-1)}} \cdot \frac{\partial v^{(t-1)}}{\partial H} = (h^{(t-2)})^T \delta^{(t-1)} \sigma'(v^{(t-1)})$$

(d)

엄밀하지는 않지만, 대략적으로 구해보면,

① $h^{(t)} = \text{Sigmoid}(\underbrace{h^{(t-1)}H}_{\substack{(1,D_h)\times(D_h,D_h) \\ \to D_h^2}} + \underbrace{e^{(t)}I}_{\substack{(1,d)\times(d,D_h) \\ \to dD_h.}} + b_1)$

$\hat{y}^{(t)} = \text{Softmax}(\underbrace{h^{(t)}U}_{\substack{(1,D_h)\times(D_h,|V|) \\ \to \boxed{D_h\cdot|V|}}} + b_2)$

확률을 구하기 위해 U와 곱하는 부분에서 load가 많이 걸린다.

$\boxed{\text{Bonus}}$

word2vec에서 negative sampling을 하듯, 모든 vocabulary |V|에 대해 확률을 구하지 않으면 load를 줄일 수 있다.

$\Rightarrow$ Forward Propagation : $O(D_h^2 + dD_h + D_h\cdot|V|)$.

② $\frac{\partial J}{\partial U} = \underbrace{(h^{(t)})^T \delta_1^{(T)}}_{\substack{\cancel{(1,D_h)\times} \\ (D_h,1)\times(1,|V|) \\ \to \boxed{D_h|V|}}}$ $\quad \frac{\partial J}{\partial h^{(t-1)}} = \underbrace{\delta_2^{(t)} H^T}_{\substack{(1,D_h)\times(D_h,D_h) \\ \to D_h^2.}}$

$\frac{\partial J}{\partial H} = \underbrace{(h^{(t-1)})^T \delta_2^{(t)}}_{\substack{(\cancel{D_h D_h}) \\ (D_h,1)\times(1,D_h) \\ \to D_h^2}}$ $\quad \frac{\partial J}{\partial L_{x^{(t)}}} = \underbrace{\delta_2^{(t)} I^T}_{\substack{(1,D_h)\times(D_h,d) \\ \to dD_h}}$ $\quad \frac{\partial J}{\partial I} = \underbrace{(L_{x^{(t)}})^T \delta_2^{(t)}}_{\substack{(d,1)\times(1,D_h) \\ \to dD_h}}$

$\to$ Backpropagation (single) : $O(|V|D_h + dD_h + D_h^2)$.

Backpropagation (τ time steps) : $O(\tau(|V|D_h + dD_h + D_h^2))$.